

TECHVISION CONSULTING INC.
Technology Advisory Services

EMERGING TECHNOLOGIES OVERVIEW

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Executive Summary

This report was prepared by the Technology Advisory Team of TechVision Consulting Inc. to give clients a clear, practical understanding of emerging technology and the specific fields most likely to reshape their operations in the near future. Drawing on the Module 1 lecture on Emerging Technologies Overview and Its Areas, the report explains what qualifies a technology as “emerging,” walks through the areas that the World Economic Forum has flagged as transformative — Augmented Reality, Artificial Intelligence, Machine Learning, and Quantum Computing — and situates these within the broader shift toward Industry 4.0. The report closes with strategic recommendations that TechVision can use when advising businesses and government agencies on technology adoption.

1. Introduction

Business leaders and government decision-makers are constantly asked to choose which technologies deserve investment and which are simply passing trends. Making that call well requires more than knowing that a technology is “new”; it requires understanding the patterns by which technologies mature, spread, and eventually reorganize entire industries. This report addresses that need directly. It is organized in four parts: first, a working definition of emerging technology grounded in the characteristics identified by researchers Rotolo, Hicks, and Martin (2015); second, a review of four technology areas that the lecture material and the World Economic Forum identify as high-impact; third, an explanation of how these technologies fit into the larger narrative of the Fourth Industrial Revolution; and fourth, a set of practical recommendations that TechVision's clients can act on.

2. Understanding Emerging Technology

2.1 What Makes a Technology “Emerging”

A common but incomplete definition treats any technology that is currently being developed, or that will be developed within the next five to ten years, as an emerging technology. The Module 1 lecture points out that this view is too narrow, because it equates emergence with simple newness. In practice, a technology can be decades old and still be considered emerging — artificial intelligence is the clearest example. AI research began in the 1950s, yet it continues to be classified as an emerging technology because it keeps being adapted to new problems and keeps generating new research questions. What actually distinguishes an emerging technology, then, is not how recently it was invented but how it behaves: whether it is still evolving, still spreading into new applications, and still reshaping the systems around it.

2.2 The Five Characteristics of Emerging Technology

To make the concept more concrete, Rotolo, Hicks, and Martin (2015) proposed five characteristics that, taken together, signal that a technology is genuinely emerging rather than merely new. TechVision uses this framework internally when screening a client's shortlist of

candidate technologies, since it separates durable, high-potential trends from short-lived novelties. Table 1 summarizes the five characteristics in plain terms.

Characteristic	Description
Radical Novelty	The technology achieves its purpose through a genuinely new operating principle, or applies an existing idea in a way that had not been tried before, rather than being a minor tweak of what already exists.
Relatively Fast Growth	Adoption accelerates quickly across several measurable fronts at once — more organizations experimenting with it, more funding flowing in, and more prototypes and products reaching the market.
Coherence	Formerly separate lines of research and technology start to converge and reinforce one another, giving the field a recognizable identity instead of remaining a loose collection of unrelated ideas.
Prominent Impact	The technology's effects reach well beyond a single application, reshaping how entire industries operate and how institutions, workers, and customers interact with one another.
Uncertainty and Ambiguity	Because the technology's long-term impact is still unfolding, its eventual uses — and even how different groups interpret its meaning — remain unsettled and open to debate.

Table 1. The Five Characteristics of an Emerging Technology (adapted from Rotolo, Hicks, & Martin, 2015)

Read together, these five traits describe a technology that is not just novel but also gaining momentum, pulling previously separate research areas into alignment, delivering effects that ripple across an entire sector, and still carrying real uncertainty about where it will ultimately lead. Any one trait on its own is not enough evidence that a technology deserves a client's investment; it is the combination that matters.

3. Key Areas of Emerging Technology

The World Economic Forum has singled out several technology areas as having the greatest potential to disrupt long-standing industry practices. The Module 1 lecture highlights four of these areas in particular. Each is summarized below and its relevance to TechVision's advisory work.

3.1 Augmented Reality (AR)

Augmented Reality layers computer-generated images, sound, text, or other digital elements directly onto a person's view of the physical world, blending the real and the virtual into a single

enhanced experience. The lecture cites several everyday examples: in-car displays that guide drivers during parking, mobile games such as Pokémon GO that place virtual creatures into real locations, and retail applications that let a shopper preview how a piece of furniture would look inside their own living room before buying it. AR is described as a core component of the Fourth Industrial Revolution and is expected to become more sophisticated and more common in everyday use.

- Retail and e-commerce: virtual try-on and product preview tools that reduce returns and boost buyer confidence.
- Field service and manufacturing: overlaying repair instructions or part locations directly onto a technician's field of view.
- Training and onboarding: simulating hazardous or expensive scenarios safely before workers encounter them for real.

3.2 Artificial Intelligence (AI) and Machine Learning

Artificial Intelligence refers to the use of machines to carry out tasks that normally require human intelligence, including learning from data, reasoning through a problem, and correcting mistakes without being explicitly reprogrammed. Machine Learning is best understood as a practical application of AI: it gives a system the ability to improve its performance through exposure to experience, in much the same way a person improves through practice. The lecture notes that machine learning already touches healthcare, financial services, marketing, and transportation. In healthcare specifically, machine learning models are being used to identify and develop candidate medicines far faster than traditional research methods allow.

- Healthcare: accelerating drug discovery and supporting diagnostic decision-making.
- Financial services: detecting fraud patterns and personalizing product recommendations.
- Transportation and logistics: optimizing routing and predictive vehicle maintenance.

3.3 Quantum Computing

Where conventional computers process information as bits that are either 0 or 1, quantum computers use qubits, which can exist in a superposition of both states simultaneously. This property allows a quantum computer to work through certain categories of problems dramatically faster than a classical machine could. The largest working quantum systems demonstrated so far come from organizations such as IBM, Google, Rigetti Computing, and IonQ, and a growing community of researchers and academics is now building the software layer needed to make the hardware useful. Once the technology matures further, it is expected to simulate physical phenomena — atoms, molecules, and novel materials — at a level of detail that is currently out of reach, which would have direct implications for pharmaceuticals, materials science, and advanced manufacturing.

4. The Fourth Industrial Revolution

4.1 From Steam Power to Smart Systems

Each major leap in production technology has historically been given its own “industrial revolution” label, because each leap fundamentally altered how people worked and lived, not just how goods were produced. Table 2 traces this progression through to the current, still-unfolding fourth stage.

Stage	Approx. Period	Defining Shift
Industry 1.0	18th century	Mechanization powered by water and steam replaced manual production methods.
Industry 2.0	19th century	Electricity enabled mass production, assembly lines, and more advanced machinery.
Industry 3.0	Mid-20th century	Computers and electronics automated production processes and introduced early digital control systems.
Industry 4.0	21st century (ongoing)	Physical and digital systems merge through IoT, AI, robotics, AR, and VR, enabling smart, interconnected operations.

Table 2. The Four Stages of the Industrial Revolution

Industry 4.0 is distinct from the stages before it because it does not introduce a single new tool the way steam power or electricity did. Instead, it fuses physical machinery with digital intelligence, so that production systems, suppliers, and customers all interact through a shared, data-driven layer. This is expected to change not only how goods are made but also the relationships between the people and machines involved in making them.

4.2 Building Blocks of Industry 4.0

The lecture identifies several technologies that function as the foundation of Industry 4.0. Figure 1 shows how they connect to and reinforce one another.

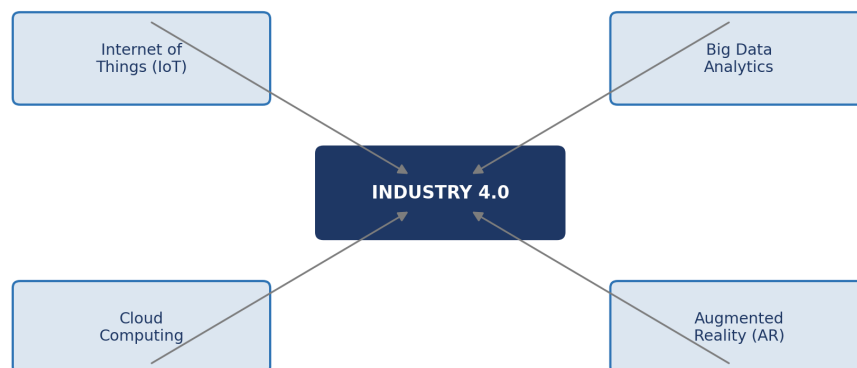


Figure 1. Interconnected Building Blocks of Industry 4.0

- Big Data Analytics – Collecting, organizing, and analyzing large volumes of data from multiple sources to uncover patterns that support real-time decisions in production and customer management.
- Internet of Things (IoT) – Connecting everyday physical objects to the internet so they can communicate and interact with each other and with central controllers, enabling decentralized, real-time responses.
- Cloud Computing – Delivering computing services over the internet, which shortens reaction times to milliseconds and makes data-driven services easier to deploy across production systems.
- Augmented Reality (AR) – Supporting tasks such as locating warehouse parts or sending repair instructions to mobile devices, giving workers real-time information at the point of need.

5. Strategic Implications and Recommendations

For TechVision's clients, the value of this material is not academic — it is a screening tool for deciding where to place limited technology budgets. The following recommendations translate the concepts above into action.

- Screen candidates against the five characteristics. Before committing budget to a technology, check it against radical novelty, growth rate, coherence, potential impact, and uncertainty. A technology that is merely “new” but fails most of these tests is a weaker bet than one that is older but still clearly emerging, such as AI.
- Prioritize AI and Machine Learning for near-term wins. Because these technologies already have proven use cases in healthcare, finance, and logistics, clients in those sectors can expect a faster and more predictable return than from earlier-stage technologies such as quantum computing.
- Treat quantum computing as a long-horizon watch item. Clients with heavy research, materials, or pharmaceutical interests should track quantum computing developments now, even if practical deployment remains several years away.
- Pilot AR in customer-facing and field-service functions. Because AR applications are relatively mature and low-risk, they are a reasonable first step for clients who are cautious about larger digital transformation efforts.
- Treat IoT, cloud computing, big data, and AR as an interconnected package, not separate purchases. Clients aiming for an Industry 4.0-style transformation should plan investments in these four areas together, since their value compounds when they are integrated rather than adopted in isolation.
- Budget for uncertainty. Because ambiguity is an inherent characteristic of emerging technology, clients should favor phased pilots with clear checkpoints over large, one-time commitments.

6. Conclusion

Emerging technology is best understood not as a list of new gadgets but as a pattern of behavior: technologies that are radically novel, growing quickly, converging with related fields, generating industry-wide impact, and still carrying real uncertainty about their ultimate direction. Augmented Reality, Artificial Intelligence and Machine Learning, and Quantum Computing each fit this pattern today, and each is closely tied to the broader shift toward Industry 4.0, in which physical and digital systems increasingly operate as one connected whole. For TechVision Consulting Inc., this framework offers a disciplined way to advise clients: rather than chasing whatever is newest, clients are better served by identifying which technologies show the full set of emerging characteristics and by sequencing investment according to how mature and interconnected those technologies already are.

The image displays two screenshots of the Quillbot website interface. The left screenshot shows the 'Detect AI' tool, which has analyzed a document and found 0% AI-generated content. It also offers a 'Refine with Paraphraser' option. The right screenshot shows the 'Writing assistant' tool, which provides suggestions for improving clarity and fluency. At the bottom, a summary bar shows scores for Grammar (97), Fluency (62), Clarity (72), and Engagement (92), with an overall average score of 83.

Left Screenshot: Detect AI

0% of text is likely AI

AI-generated 0%
Human-written & AI-refined 0%
Human-written 100%

Model v7.1.0

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Want more authentic text? [Refine with Paraphraser](#)

Right Screenshot: Writing assistant

Grammar corrections and writing suggestions

Clarity • Consider active voice Premium

Fluency • Improve fluency Premium

Summary Scores:

- Grammar: 97
- Fluency: 62
- Clarity: 72
- Engagement: 92

Average: 83

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